

Changes Around Us

MCQ

1. Change is said to be constant because

- a) Objects never move
- b) Everything remains same
- c) Changes occur continuously around us
- d) Only living things change

Answer: c) Changes occur continuously around us

2. Which of the following is a natural change?

- a) Cutting paper
- b) Burning wood
- c) Day changing into night
- d) Cooking food

Answer: c) Day changing into night

3. Growth of hair and nails is an example of

- a) Fast change
- b) Periodic change
- c) Slow change
- d) Reversible change

Answer: c) Slow change

4. A change that occurs over a long period of time is

- a) Sudden change
- b) Slow change
- c) Periodic change
- d) Chemical change

Answer: b) Slow change

5. Which of the following involves a change in state?

- a) Rusting of iron
- b) Burning of paper

- c) Melting of ice
- d) Curdling of milk

Answer: c) Melting of ice

6. Change in shape without forming a new substance is

- a) Chemical change
- b) Physical change
- c) Permanent change
- d) Periodic change

Answer: b) Physical change

7. Which of the following is a physical change?

- a) Rusting
- b) Burning
- c) Fermentation
- d) Boiling of water

Answer: d) Boiling of water

8. A change in which no new substance is formed is called

- a) Chemical change
- b) Irreversible change
- c) Physical change
- d) Periodic change

Answer: c) Physical change

9. Which change is reversible?

- a) Burning of wood
- b) Rusting of iron
- c) Melting of ice
- d) Curdling of milk

Answer: c) Melting of ice

10. Rusting of iron is a

- a) Physical change
- b) Reversible change

c) Temporary change

d) Chemical change

Answer: d) Chemical change

11. Which change is permanent?

a) Stretching rubber

b) Melting ice

c) Burning paper

d) Freezing water

Answer: c) Burning paper

12. A new substance is formed during

a) Freezing

b) Melting

c) Chemical change

d) Physical change

Answer: c) Chemical change

13. Which of the following is an irreversible change?

a) Boiling water

b) Freezing water

c) Rusting of iron

d) Melting wax

Answer: c) Rusting of iron

14. Which state of matter has definite shape and volume?

a) Liquid

b) Gas

c) Solid

d) Plasma

Answer: c) Solid

15. Molecules are tightly packed in

a) Liquids

b) Gases

- c) Solids
- d) Vapours

Answer: c) Solids

16. Which state has no definite shape but definite volume?

- a) Solid
- b) Liquid
- c) Gas
- d) Ice

Answer: b) Liquid

17. Gases have

- a) Strong intermolecular force
- b) Definite shape
- c) Very weak intermolecular force
- d) Fixed volume

Answer: c) Very weak intermolecular force

18. Matter is defined as anything that

- a) Can be seen
- b) Is solid
- c) Has mass and occupies space
- d) Is useful

Answer: c) Has mass and occupies space

19. Heating generally causes matter to

- a) Shrink
- b) Expand
- c) Freeze
- d) Condense

Answer: b) Expand

20. Cooling generally causes matter to

- a) Expand
- b) Melt

c) Contract

d) Boil

Answer: c) Contract

21. Melting is the change from

a) Liquid to solid

b) Gas to liquid

c) Solid to liquid

d) Gas to solid

Answer: c) Solid to liquid

22. Melting is an example of

a) Chemical change

b) Exothermic process

c) Physical change

d) Irreversible change

Answer: c) Physical change

23. Melting absorbs heat, so it is a

a) Exothermic process

b) Endothermic process

c) Chemical reaction

d) Neutral process

Answer: b) Endothermic process

24. Evaporation is a change from

a) Solid to liquid

b) Liquid to gas

c) Gas to liquid

d) Solid to gas

Answer: b) Liquid to gas

25. Evaporation occurs only at

a) High pressure

b) Entire liquid

- c) Surface of liquid
- d) Low temperature

Answer: c) Surface of liquid

26. Which factor increases evaporation?

- a) High humidity
- b) Low temperature
- c) Larger surface area
- d) Still air

Answer: c) Larger surface area

27. Boiling occurs at

- a) Liquid surface only
- b) Entire mass of liquid
- c) Low temperature
- d) Low pressure only

Answer: b) Entire mass of liquid

28. Freezing is the change from

- a) Gas to liquid
- b) Liquid to solid
- c) Solid to gas
- d) Liquid to gas

Answer: b) Liquid to solid

29. Freezing releases heat, so it is

- a) Endothermic
- b) Chemical
- c) Exothermic
- d) Neutral

Answer: c) Exothermic

30. Condensation is opposite of

- a) Melting
- b) Freezing

c) Evaporation

d) Sublimation

Answer: c) Evaporation

31. Condensation changes

a) Solid to gas

b) Gas to liquid

c) Liquid to solid

d) Solid to liquid

Answer: b) Gas to liquid

32. Sublimation is the change from

a) Solid to liquid

b) Liquid to gas

c) Solid to gas

d) Gas to solid

Answer: c) Solid to gas

33. Which substance shows sublimation?

a) Ice

b) Salt

c) Camphor

d) Sugar

Answer: c) Camphor

34. Crystallisation is used to

a) Burn substances

b) Purify solids

c) Melt solids

d) Compress gases

Answer: b) Purify solids

35. Crystallisation is a

a) Chemical change

b) Irreversible change

c) Physical change

d) Nuclear change

Answer: c) Physical change

36. Chemical change results in

a) Same substance

b) Temporary effect

c) New substance

d) Only colour change

Answer: c) New substance

37. Which is an indicator of chemical change?

a) Shape change

b) Size change

c) Colour change

d) Position change

Answer: c) Colour change

38. Formation of precipitate indicates

a) Physical change

b) Chemical change

c) Reversible change

d) Periodic change

Answer: b) Chemical change

39. Rusting requires

a) Water only

b) Oxygen only

c) Air and moisture

d) Heat only

Answer: c) Air and moisture

40. Galvanisation prevents rusting by coating iron with

a) Copper

b) Aluminium

c) Zinc

d) Silver

Answer: c) Zinc

41. Burning of wood needs

a) Oxygen, fuel, ignition temperature

b) Only fuel

c) Only air

d) Only heat

Answer: a) Oxygen, fuel, ignition temperature

42. Ignition temperature is

a) Maximum temperature

b) Lowest temperature at which substance catches fire

c) Room temperature

d) Melting point

Answer: b) Lowest temperature at which substance catches fire

43. Paper burns faster than wood because

a) Paper is heavier

b) Paper has lower ignition temperature

c) Wood has oxygen

d) Paper is wet

Answer: b) Paper has lower ignition temperature

44. Magnesium ribbon burns with

a) Yellow flame

b) Blue flame

c) Bright white light

d) Red flame

Answer: c) Bright white light

45. Magnesium burns to form

a) Magnesium sulphate

b) Magnesium oxide

c) Magnesium chloride

d) Magnesium nitrate

Answer: b) Magnesium oxide

46. Curdling of milk is a

a) Physical change

b) Reversible change

c) Chemical change

d) Periodic change

Answer: c) Chemical change

47. Milk turns into curd due to

a) Heat

b) Oxygen

c) Lactobacillus bacteria

d) Water

Answer: c) Lactobacillus bacteria

48. Fermentation occurs in the absence of

a) Water

b) Heat

c) Air

d) Sugar

Answer: c) Air

49. Yeast converts sugar into

a) Acid

b) Salt

c) Alcohol and carbon dioxide

d) Water

Answer: c) Alcohol and carbon dioxide

50. Fermentation is an

a) Reversible process

b) Physical change

c) Chemical change

d) Periodic change

Answer: c) Chemical change

51. Catalyst is a substance that

a) Takes part in reaction

b) Slows reaction

c) Speeds reaction without being used

d) Stops reaction

Answer: c) Speeds reaction without being used

52. Baking soda is

a) Acid

b) Base

c) Salt

d) Neutral

Answer: b) Base

53. Chemical formula of baking soda is

a) NaCl

b) NaOH

c) NaHCO_3

d) HCl

Answer: c) NaHCO_3

54. Lemon contains

a) Acetic acid

b) Citric acid

c) Lactic acid

d) Hydrochloric acid

Answer: b) Citric acid

55. Reaction of acid and base is called

a) Combustion

b) Decomposition

c) Neutralisation

d) Oxidation

Answer: c) Neutralisation

56. CO₂ bubbles indicate

a) Physical change

b) Chemical change

c) Periodic change

d) Reversible change

Answer: b) Chemical change

57. Endothermic process

a) Releases heat

b) Absorbs heat

c) Has no energy change

d) Produces light

Answer: b) Absorbs heat

58. Which is an endothermic process?

a) Freezing

b) Condensation

c) Melting

d) Deposition

Answer: c) Melting

59. Exothermic process

a) Absorbs heat

b) Releases heat

c) Absorbs light

d) Produces no energy

Answer: b) Releases heat

60. Which is an exothermic process?

a) Evaporation

b) Sublimation

c) Freezing

d) Melting

Answer: c) Freezing

61. Periodic changes occur

a) Suddenly

b) Randomly

c) At regular intervals

d) Once in lifetime

Answer: c) At regular intervals

62. Day and night is an example of

a) Non-periodic change

b) Chemical change

c) Periodic change

d) Slow change

Answer: c) Periodic change

63. Seasonal changes are

a) Irreversible

b) Non-periodic

c) Periodic

d) Chemical

Answer: c) Periodic

64. Non-periodic changes occur

a) Regularly

b) At fixed time

c) Randomly

d) Daily

Answer: c) Randomly

65. Earthquake is a

a) Periodic change

b) Physical change

c) Non-periodic change

d) Chemical change

Answer: c) Non-periodic change

66. Growing crops is an example of

a) Chemical change

b) Periodic change

c) Non-periodic change

d) Sudden change

Answer: b) Periodic change

67. Falling of leaves is

a) Chemical

b) Periodic

c) Non-periodic

d) Reversible

Answer: c) Non-periodic

68. Stretching rubber band is

a) Chemical change

b) Reversible physical change

c) Irreversible change

d) Periodic change

Answer: b) Reversible physical change

69. Cutting paper is

a) Reversible

b) Chemical

c) Irreversible physical change

d) Periodic

Answer: c) Irreversible physical change

70. Boiling water is

a) Chemical change

b) Irreversible

c) Physical and reversible

d) Permanent

Answer: c) Physical and reversible

71. Which change involves smell?

a) Melting ice

b) Burning paper

c) Stretching rubber

d) Freezing water

Answer: b) Burning paper

72. Colour change usually indicates

a) Physical change

b) Chemical change

c) Periodic change

d) Temporary change

Answer: b) Chemical change

73. Photosynthesis is a chemical change because

a) Light is used

b) New substances are formed

c) It is slow

d) It is reversible

Answer: b) New substances are formed

74. Pressure cooking rice is a

a) Physical change

b) Chemical change

c) Reversible change

d) Periodic change

Answer: b) Chemical change

75. Electrolysis of water produces

a) Steam only

b) Hydrogen and oxygen

c) Ice and vapour

d) Acid and base

Answer: b) Hydrogen and oxygen

76. Electricity causes

a) Only physical change

b) No change

c) Chemical change

d) Periodic change

Answer: c) Chemical change

77. Heat during burning is

a) Absorbed

b) Released

c) Unchanged

d) Neutral

Answer: b) Released

78. Cracking of firecrackers shows

a) Only sound

b) Physical change

c) Chemical change

d) Periodic change

Answer: c) Chemical change

79. Spoiling of food is indicated by

a) Colour only

b) Taste and smell

c) Shape change

d) Size change

Answer: b) Taste and smell

80. Which change involves both physical and chemical processes?

a) Cooking food

b) Melting ice

- c) Freezing water
- d) Stretching rubber

Answer: a) Cooking food

81. Water droplets on cold bottle are due to

- a) Evaporation
- b) Freezing
- c) Condensation
- d) Sublimation

Answer: c) Condensation

82. Ice crystals form by

- a) Evaporation
- b) Sublimation
- c) Deposition
- d) Melting

Answer: c) Deposition

83. Deposition is change from

- a) Gas to solid
- b) Liquid to gas
- c) Solid to liquid
- d) Liquid to solid

Answer: a) Gas to solid

84. Increase in wind speed causes evaporation to

- a) Decrease
- b) Stop
- c) Increase
- d) Remain same

Answer: c) Increase

85. High humidity causes evaporation to

- a) Increase
- b) Stop

c) Decrease

d) Double

Answer: c) Decrease

86. Which has maximum intermolecular space?

a) Solid

b) Liquid

c) Gas

d) Ice

Answer: c) Gas

87. Which has strongest intermolecular force?

a) Gas

b) Liquid

c) Solid

d) Vapour

Answer: c) Solid

88. Change in position is a

a) Chemical change

b) Physical change

c) Periodic change

d) Permanent change

Answer: b) Physical change

89. Which is NOT a chemical change?

a) Rusting

b) Burning

c) Melting

d) Fermentation

Answer: c) Melting

90. Chemical changes are generally

a) Temporary

b) Reversible

c) Permanent

d) Periodic

Answer: c) Permanent

91. Physical changes are generally

a) Irreversible

b) Permanent

c) Temporary

d) Chemical

Answer: c) Temporary

92. Salt obtained from seawater uses

a) Sublimation

b) Crystallisation

c) Burning

d) Condensation

Answer: b) Crystallisation

93. Arrangement of particles in crystals is

a) Random

b) Loosely packed

c) Well defined

d) Scattered

Answer: c) Well defined

94. Which change involves heat absorption?

a) Freezing

b) Condensation

c) Melting

d) Deposition

Answer: c) Melting

95. Which change involves heat release?

a) Evaporation

b) Sublimation

c) Freezing

d) Melting

Answer: c) Freezing

96. Running between wickets is

a) Chemical change

b) Periodic change

c) Non-periodic change

d) Reversible change

Answer: c) Non-periodic change

97. Dancing movements are

a) Periodic

b) Non-periodic

c) Chemical

d) Permanent

Answer: b) Non-periodic

98. Clock bell ringing every hour is

a) Non-periodic

b) Chemical

c) Periodic

d) Physical

Answer: c) Periodic

99. Which change does NOT involve energy?

a) Burning

b) Melting

c) Rusting

d) None

Answer: d) None

100. Changes help us understand

a) Only chemistry

b) Only physics

c) Nature and matter behavior

d) Mathematics

Answer: c) Nature and matter behavior